

Aahil Khan

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SKILLS

- **Languages:** Python, TypeScript, JavaScript
- **Backend & APIs:** Node.js, Express.js, Fastify, FastAPI, Flask, REST APIs
- **Frontend & UI:** React.js, Next.js, Tailwind CSS, Shadcn
- **Databases & Storage:** PostgreSQL, MySQL, Redis, Prisma
- **AI Systems:** RAG pipelines, semantic search, vector retrieval, structured outputs, embeddings, LLM APIs
- **Infrastructure & DevOps:** Docker, Nginx, CI/CD (GitHub Actions)
- **Cloud & Networking:** AWS (EC2, Lambda, API Gateway, DynamoDB, S3)
- **Testing:** Jest, Vitest, Postman
- **AI Tooling:** Cursor, ChatGPT, Claude, Gemini, GitHub Copilot

PROFESSIONAL EXPERIENCE

Backend Engineer — Datesheet Generator

Flask, MySQL, Python | Thapar University

Patiala, India

Mar 2025 – July 2025

- Re-architected the system from session-based state to a persistent MySQL-backed workflow.
- Implemented secure login, multi-user support, and savepoint functionality for partial work resumption.
- Improved datesheet generation workflows, reducing scheduling errors by 40%.
- Integrating with external university systems (Oracle PeopleSoft) to automate structured data ingestion and export.

Team Lead / Full-Stack Engineer — Thapar EduTube

Node.js, Express, PostgreSQL FTS, Redis, Docker, AWS EC2 | Thapar University

Patiala, India

Oct 2024 – Jan 2025

- Led a team of 3 engineers to build a lecture archival platform serving 10,000+ university students.
- Designed backend APIs for lecture ingestion, metadata management, full-text search, and access control.
- Built full-stack admin and teacher dashboards for lecture uploads, course management, and workflow customization.
- Implemented search functionality using PostgreSQL Full-Text Search after evaluating Elasticsearch, optimizing for latency and operational simplicity.
- Deployed and operated production services on AWS EC2 using Docker, ensuring reliability for daily academic use.

SELECTED PROJECTS

FlowSync — AI Project Memory for Coding Agents

Amazon Bedrock, AWS Lambda, API Gateway, DynamoDB, S3, Next.js, VS Code Extension, MCP, TypeScript, Python

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- Built an AI project memory system giving coding agents persistent context via the Model Context Protocol (MCP).
- Developed a VS Code extension and MCP server for context logging, project search, recent changes, and branch-aware RAG pipeline with vector retrieval.
- Built an event-driven async AWS pipeline extracting decisions, risks, and tasks from git diffs using Bedrock models.
- Implemented a branch-aware RAG system with DynamoDB caching achieving 1.26s median push-to-search latency (p95: 1.60s).

Konta — Context-Aware Browsing System (Samsung PRISM Web Agent Hackathon Winner)

Chrome Extension | TypeScript, React, IndexedDB, Transformers.js, ONNX

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- Built a local-first Chrome extension that converts browsing activity into an on-device knowledge base.
- Implemented on-device semantic embeddings and a multi-stage retrieval pipeline for private, low-latency search.
- Designed a knowledge graph linking system connecting pages using semantic, temporal, and contextual relationships.
- Engineered sessionization logic with context boundary detection to prevent unrelated browsing tasks from merging.

Portana — AI Powered Interactive Portfolio

Fastify, TypeScript, Qdrant, Docker, n8n, Umami Analytics

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- Built an AI-driven portfolio platform with conversational project exploration and dynamic project insights.
- Developed modular API routes enabling real-time rendering and smoother user interactions.
- Added n8n-based auto-sync workflows for GitHub/LinkedIn to keep content continuously updated.
- Integrated analytics and monitoring to track usage and system behavior.

GINA — Grounded Insights from Natural Language Analytics

Next.js, TypeScript, Fastify, PostgreSQL, Supabase Auth, SSE

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- Built a multi model NL→SQL pipeline for conversational data analytics with structured outputs.
- Implemented SSE streaming with fast path routing (snapshot + cache restore) improving latency and transparency.
- Designed context aware multi turn query handling using structured conversation state for grounded follow ups.
- Added semantic schema correction loops and failure tolerant API/SSE error handling for reliable analytics workflows.

ACHIEVEMENTS

Winner — Samsung PRISM Web Agent Hackathon

2025

Innovation Award — Agentic AI Hackathon, Ulster University (UK)

2025

EDUCATION

Thapar Institute of Engineering Technology

CGPA: 9.23

Bachelor of Engineering in Computer Engineering | Aug 2023 – Present

Indian Institute of Technology Madras

CGPA: 8.34

Diploma in Programming | Sept 2023 – Present